COMP1787 Requirements Management

Lecture 1: Introduction to the Module - History of Software and System Development

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Introduction to the Module

- *This module will introduce you to key issues associated with management of requirements.*
- *Quality Information systems are critical to the success of today's organisations.*
- *Underpinning the development of such quality systems is the successful management of the requirements engineering cycle.*

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Learning Outcomes

Analyse and compare current approaches to requirements management within a development environment

Assess the impact of stakeholders and organisational culture on the development of effective requirements and system development

Relate issues associated with risk, quality, and Legal/Social/Ethical/Professional (LSEPI) to a practical scenario

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Teaching Schedule

Tutorials:

- COMP1787/Tutorial/01-01. 12:00-1:00pm QM168 Aditi Rawal
- COMP1787/Tutorial/01-02 - 12:00-1:00pm QA075 Irfan Chishti

- You are expected to attend the tutorial allocated to you.

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Assessment

Report - 100% weighting (3000 words)

Evaluating the case study-based provided

The coursework submission deadline is Tuesday 25 March 2025.

The overall pass mark for this module is 40%.

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Background to Systems Development

For a system to be considered *successful* it needs to essentially do two things:

Meet the users' expectations (Quality target)	Be delivered on time and to budget (Productivity target)

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Quality Targets

The system meets the requirements specification (i.e., it serves the purpose for which it was originally intended)

these targets relate to

an understanding of the business area	the user requirements	the organisational constraints	other environmental factors

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Quality issues resulting in unsuccessful systems

- The wrong problem has been identified
- Neglect of wider organisational issues
- Incorrect analysis
- System developed for the wrong reasons

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Productivity targets

The system is delivered **on time** and **within budget.**

These targets relate to:

successful management of the project,	from inception to implementation

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Productivity issues resulting in unsuccessful systems

- Users change their minds
- External events may occur that change the requirements
- Implementation may not be feasible
- Another general cause for an unsuccessful system may be poor project control.

Systems Failures (1)

- Unsuccessful systems are not new.
- The complexity of a system does not necessarily relate to its chances of success or failure.
- Central criticism of information systems has been (and still is) that:
  - *many systems do not do what their users require and thus fall into disuse.*

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Systems failures (2)

Early problems of quality and productivity discussed in the NATO SE Conference (1968) in Garmisch and then in 1969 in Rome

- getting requirements right,
- designing adequate architectures,
- doing implementation effectively and correctly,
- verifying the quality of the result, and
- maintaining software systems with targeted functionality and high code quality over long time periods and to further develop them.

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Systems failures (3)

- Patterned continued well into the next 2 decades
- Department of Trade and Industry (DTI) report (1985) found that:

  - *time-scale overruns occurred in 66% of projects.*
  - *55% were over budget.*

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Systems failures (4)

- KPMG (1990) quoted:

  "Runaway systems over 30% of all major projects"

- *Major effects of these systems were:*
  - *Loss of time*
  - *reduction in staff morale*
  - *loss of money*
  - *customer satisfaction*
  - *a negative market image*

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Systems failures (6)

More failures:

- KPMG report for the 2005 survey:
  - 49% of participants experienced at least one project failure in the last year.
- KPMG report for the 2010 survey:
  - 70% of participants experienced at least one project failure in the last year.
- KPMG report for the 2019 survey:
  - 19% Of organisations deliver successful projects, at least most of the time
  - 35% of organisations are likely to deliver projects that are on budget

(Source: https://www.ipma.world/assets/PM-Survey-FullReport-2019-FINAL.pdf)

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Systems failures (6)

- Many high-profile failures published in the 1990s.
- Many of these systems were in the public sector

  - Social security payments system (Pathway)
    - *blighted from the outset*
    - *cost £1billion taxpayers money (written off in the end) according to National Audit Office.*
    - *Should have been delivered in 1997*
    - *Finally scrapped*

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Problems with the Pathways Project

- Stakeholder disagreements
- Efforts to save time and money at the beginning of the project meant there was insufficient preparatory work to get specifications right.
- The result dramatically increased costs further down the line.
- The precise nature and specification of the system's security and reporting procedures remained unresolved when the card project was cancelled three years later
- There was no shared, open approach to risk management.

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Systems failures (7)

More system failures through the 1990s

- London Ambulance system
  - Classic example of poor analysis and design
  - Complete misunderstanding of the environment and *Human Activity Systems*
  - Loss of lives and £7.5 million

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London Ambulance System Problems

- Ambiguity over consultation of ambulance crews for development of original requirements.
- Lack of voice contact exacerbated “them and us”.
- Poor interface between crews, MDTs & the system
- Numerous technical problems with the system, including:
- Failure to identify all duplicated calls;
- Lack of prioritisation of exception messages;
- Exception messages and awaiting attention queues scroll off top of screen.
- Going ‘pan London’ rather than operating in 3 divisions

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Systems failures (8)

The Millennium has also seen many system failures

- Child Support Act (CSA) system
  - Went live in March 2003
  - 2 years late and didn't work properly
  - Thousands of parents were waiting for files to be moved to new system
  - After awarding the system to EDS the CSA dept gave them over 2,500 changes to the system design
  - The National Audit Office (NAO) described the 2003 reforms as an "unsuccessful attempt" to deliver the policy behind the 1993 scheme.
  - It noted that 923,000 cases remained on the more complex 1993 scheme and that it cost £0.70 to collect every £1 from a paying parent in 2004/5.

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Systems failures (9)

- Libra magistrates court system
  - The cost of the Libra project to provide a national system for 385 magistrates courts soared from £146m to £390m
  - 11 years for project delivery estimated
  - Supposed to deliver *a unified case management system to Magistrates courts in England & Wales*
  - Started in 1992. In September 2004 decided that *it is unfit for purpose*

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2000 – 2015

- Methodologies
  - RAD , XP for smaller projects
  - Larger still used used heavily documented methodologies

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System failures (10)

NHS UK's National Programme for IT (NPfIT)

- 12.4 billion programme
- Plagued with problems from outset
- Systems didn't do what the users wanted it to do (little consultation)
- Emphasis was placed infrastructure and not on users needs
- Too many contractors (making money for themselves)
- 22nd September 2011
  - The Cabinet Office's Major Projects Authority reported that "I*t has not and cannot deliver to its original intent*".

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NHS UK's National Programme for IT (NPfIT) cont.

Unrealistic timetable	No time to engage with users and privacy campaigners	Inadequate preliminary work
Failure to check progress against expectations	Failure to test systems	No risks management of big IT projects
Failure to recognize that the longer the project takes, the more likely it is to be overtaken by new technology	Too large for the leadership to manage competently	Confidentiality issues

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Systems failures (11)

National Offender Management Information System (NOMIS)

- Joint project by Home Office, the Ministry of Justice launched in 2004
- Developed by EDS later acquired by HP Enterprise Services
- Purpose to provide a single database of offenders, to help with prisons and probation services management.
- Loss of £690 million taxpayers' money

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Cont…

- National Audit Office (2009) found:
  - there was so little control of C-Nomis that the government spent £161m on systems without anyone knowing how or exactly on what.
  - The system development was guilty of ‘basic’ project management failures

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System failures (12)

FireControl

- 20th September 2011: “The doomed attempt to streamline fire control services in England is one of the worst cases of project failure’ the Public Accounts Committee has seen” its chair said today.
- Estimated costs soared from £120 million to over £469 million since 2004.

Source: https://www.publicfinance.co.uk/news/2011/09/firecontrol-one-worst-cases-project-failure-says-pac

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Cont ....

- Main reasons for its failure:
  - the project was launched too quickly
  - decisions taken before a business case or procurement strategy had been developed and tested.
  - There was an over-reliance on external consultants
  - Local fire and rescue services resisted the changes, preferring their existing systems.
  - Ineffective oversight and coordination among stakeholders.
  - A high turnover of senior managers, none of whom have been held accountable for Fire Control's failure.

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Systems Failure (13)

COVID Test and Trace (2020)

- A total of £10bn was allocated by the government to spend on a national test and trace system, with private contractor Serco handed a leading role.
- GPs and primary care teams should have played a key role in the UK's efforts to test, track and trace coronavirus, according to a leading global health expert who has labelled the current system 'a disaster and a national shame'.
- Primary care had been 'bypassed totally' in the government's strategy to tackle the virus

(source:https://www.gponline.com/failure-involve-gps-covid-19-test-trace-a-disaster-national-shame/article/1698043

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Systems Failure (14)

- Post Office's Horizon system
  - The Horizon IT scandal, frequently called the largest miscarriage of justice in British history
  - Intended to streamline operations to replace paper-based tills.
  - The system had a networked structure and a backend that connected the entire Post Office system.
  - Horizon is software that was originally developed by UK software company ICL, which was acquired by Japanese IT giant Fujitsu in 1998.
  - The system has an electronic point of sale service (EPOSS) that enables subpostmasters and branch workers to key in sales on a touchscreen, with the accounts being automated in the background. At the time of its roll-out in 1999/2000, it was the biggest non-military IT project in Europe.

Source Acknowlegment: https://www.computerweekly.com/feature/Post-Office-Horizon-scandal-explained-everything-you-need-to-know

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Post Office’s Horizon system cont.

- Intended to streamline operations, the system had a networked structure and a backend that connected the entire Post Office system.
- The system faced significant issues, with trials in 1999 revealing severe difficulties for sub postmasters.
- David McDonnell, a member of the development team, highlighted the team's varying levels of competency, with some incapable of producing professional code.
- By 2001, hundreds of bugs were identified in the Horizon system, leading to operational problems.

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cont.

- Dalmellington bug created a £24,000 discrepancy, which the Post Office tried to hold the post office operator responsible for.
- Another bug, called the Callendar Square bug – again named after the first branch found to have been affected by it – created duplicate transactions due to an error in the database underpinning the system: despite being clear duplicates, the post office operator was again held responsible for the errors.

- Despite functionality claims, the Horizon system fell short of acceptable standards.
  - In 2015, the Post Office falsely stated that there was no capability to edit or remove transaction data in Horizon.
  - Four years later, during a court case, the Post Office admitted the earlier statement was untrue.
  - Fujitsu, the system's maker, had the ability to remotely access branch accounts with unrestricted access, contradicting previous denials.

Source: Hern, A (2024) How the Post Office's Horizon system failed: a technical breakdown. The Guardian.

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Should IT systems still fail?

- We have highly educated development staff…..
- We have IT literate users
- We have many different methodologies to support the life cycle across a wide range of different applications
- We have tried and tested technologies and development platforms

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- According to Gartner, around 80% of IT projects are considered failures by the business, resulting in cost overruns, missed deadlines, and unmet expectations. (2023)

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Problems with Systems Development?

Each organisation is unique with its own needs and problems.

Each set of end-users is unique with its own aspirations and concerns.

Any development framework/method, can only be used as a guide.

Developers still often focus too much upon the software development and not upon the requirements analysis

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Organisational Factors

- Situational factors: Top-down “Vanity” projects
- Stakeholder Engagement
- Organisational Issues: Balancing Risk and Reward
- Development methods currently used
- Checks and Balances: Course Correction

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Human Factors in Systems Development

Modern methodologies stress technological development but ignore human factors.

Role of end-user is significant in successful development of an information system.

Role of analyst is not only to elicit information system requirements but also to analyse the human environment to be supported by the system

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Key Issues

- Requirements are decided in silos
- Not solving a real user need
- Failure to revisit estimates
- Shifting focus
- Scope creep
- Unexpected twists
- Missing expertise
- Lack of stakeholder consultation and engagement.
- Overambitious goals without realistic timelines.
- Insufficient testing and phased rollouts.
- Weak project governance and oversight.

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In conclusion

Understanding the history of systems development failures

Focus tends to be on technology and software development and not on end users and their requirements

A successful system meets two targets:

- Quality target
- Productivity target

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